Experiment No.: 01

Title: Implement Box Plot using Data Set

Batch: B3 **Roll No.: 16014223083** **Experiment No.: 0 1**

Aim: To visualize the distribution of a data set using a **box plot** and analyze **median, quartiles, interquartile range (IQR), spread, and outliers**.

Resources needed: VSCode, Python, Google Colab, Jupyter Notebook

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Pre Requisite: Data Science, Machine Learning

Theory:

A **box plot** (also called a box-and-whisker plot) is a graphical method to display the distribution of numerical data through five summary statistics:

- Minimum
- First Quartile (Q1)
- Median (Q2)
- Third Quartile (Q3)
- Maximum

The **box** represents the **interquartile range (IQR = Q3 − Q1)**.
The **whiskers** extend to the smallest and largest values within 1.5 × IQR.
Any value outside this range is considered an **outlier**.

Advantages of Box Plot

1. Detecting outliers
2. Comparing distributions
3. Understanding data spread and skewness

Algorithm / Procedure

1. Collect the numerical data set.
2. Arrange the data in ascending order.
3. Calculate:
   - Q1 (25th percentile)
   - Q2 (Median)
   - Q3 (75th percentile)
4. Compute Interquartile Range:
   IQR = Q3 − Q1
5. Determine lower and upper bounds:
   - Lower Bound = Q1 − 1.5 × IQR
   - Upper Bound = Q3 + 1.5 × IQR
6. Identify outliers beyond the bounds.

7. Draw the box plot using the selected tool.
8. Interpret the plot.

Data Set

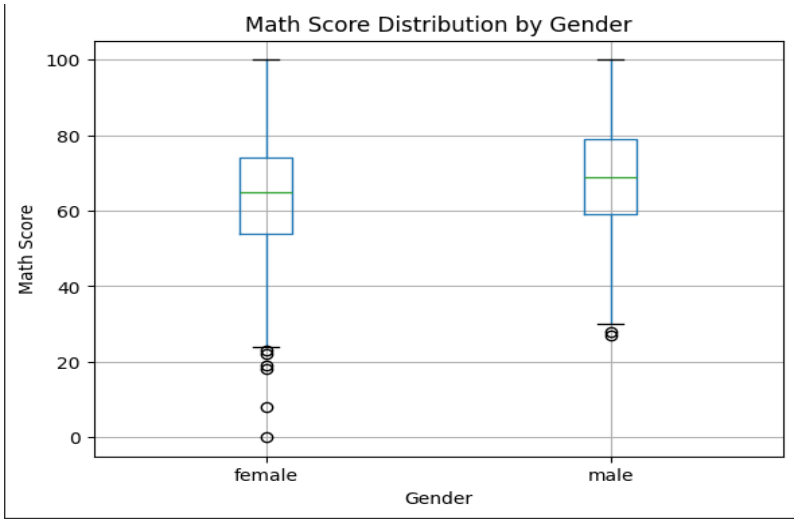
Kindly take appropriate Data set and mention the reference.
https://www.kaggle.com/datasets/spscientist/students-performance-in-exams

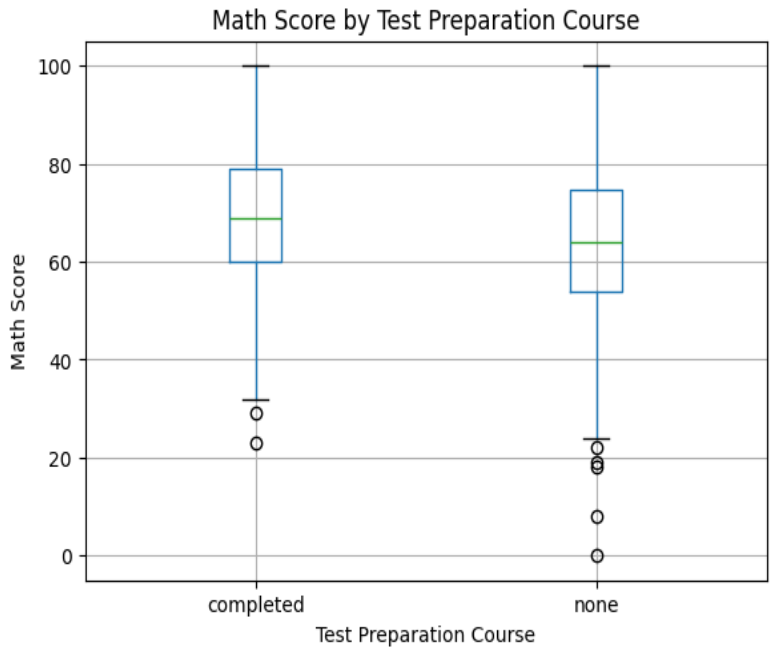
```
import pandas as pd
import matplotlib.pyplot as plt
data.boxplot(column='math score', by='gender')
plt.title("Math Score Distribution by Gender")
plt.suptitle("")
plt.xlabel("Gender")
plt.ylabel("Math Score")
plt.show()
```

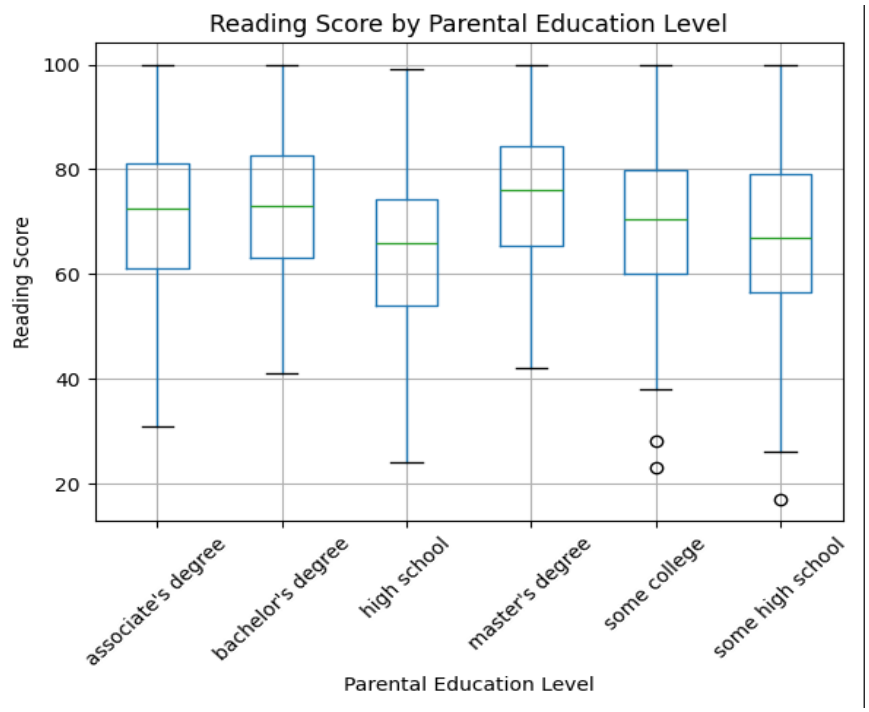
```
data.boxplot(column='math score', by='test preparation course')
plt.title("Math Score by Test Preparation Course")
plt.suptitle("")
plt.xlabel("Test Preparation Course")
plt.ylabel("Math Score")
plt.show()
```

```
data.boxplot(column='reading score', by='parental level of education', rot=45)
plt.title("Reading Score by Parental Education Level")
plt.suptitle("")
plt.xlabel("Parental Education Level")
plt.ylabel("Reading Score")
plt.show()
```

Output:







Outcomes:

CO1: Understand the basic concepts of data analytics

Conclusion: (Conclusion to be based on the objectives and outcomes achieved)

In this experiment, we successfully visualized the distribution of student performance data using box plots. The visual analysis revealed that students who completed the test preparation course generally scored higher (higher median and upper quartiles) compared to those who did not. Additionally, outliers were effectively detected as individual points falling outside the whiskers

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Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:
Websites:

- “Box Plot (Box-and-Whisker Plot),” *Wikipedia*. https://en.wikipedia.org/wiki/Box_plot
- “Box Plot,” *Investopedia*. https://www.investopedia.com/terms/b/boxplot.asp